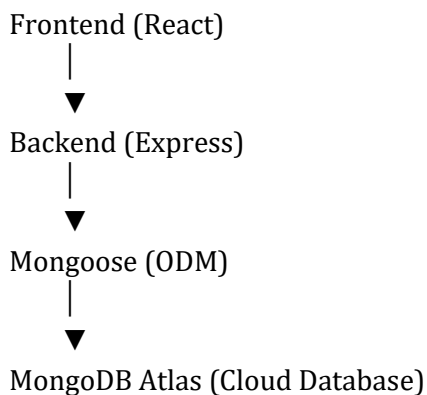


PHASE 2 - MongoDB Atlas Setup & Backend Connection (Detailed Guide)

Project: Bitto Website (MERN Stack)

Objective: Connect the Express backend to MongoDB Atlas, understand every step, troubleshoot errors, and verify successful communication.

Architecture



Step 1 - Create MongoDB Atlas Account

Visit <https://cloud.mongodb.com> and sign in. Atlas is MongoDB's managed cloud database service. We use Atlas because we don't need to install MongoDB locally and it is the same type of managed service used in many real projects.

Step 2 - Create Free Cluster

Create an M0 Free Cluster on AWS. Choose the nearest region (Hyderabad) to reduce latency. The cluster is the actual MongoDB server running in the cloud.

Step 3 - Database User

Go to Security → Database Access → Add Database User. Create user 'shreyanshs713_db_user'. This username/password is used by the backend to authenticate with Atlas. Without it Atlas returns 'Authentication Failed'.

Step 4 - Network Access

Go to Security → Network Access → Add Current IP. Atlas blocks unknown IPs by default for security. Only whitelisted IP addresses can connect.

Step 5 - Connection String

Cluster → Connect → Drivers → Node.js. Atlas provides a MongoDB URI. Every part of the URI has a meaning:

mongodb+srv:// = protocol
username = database user
password = database password
cluster.mongodb.net = server address

/bitto_catalog = database name
?appName=bitu-Cluster = application metadata.

Step 6 - Environment Variables

Create backend/.env. Store PORT, MONGO_URI, JWT_SECRET and other secrets. Secrets should never be hardcoded or pushed to GitHub. The .env file is ignored using .gitignore.

Step 7 - Mongoose Connection

db.js uses await mongoose.connect(process.env.MONGO_URI). 'await' pauses execution until the database connection succeeds. If connection fails, process.exit(1) stops the server because running an API without a database is unsafe.

Step 8 - Server Startup

server.js first loads dotenv, then connects to MongoDB, and only after a successful connection starts Express on port 5000. This guarantees the API starts only when the database is available.

Step 9 - Health API

Open <http://localhost:5000/api/health>. Successful JSON response confirms that Express is running correctly.

Commands Used

Command	Purpose	Explanation
npm install mongoose	Install ODM	Installs Mongoose library for MongoDB.
npm run dev	Run backend	Starts backend using Nodemon.
node -v	Check Node	Displays installed Node.js version.
npm list mongoose	Verify package	Shows installed Mongoose version.
npm list mongodb	Verify driver	Shows MongoDB driver version.
nslookup -type=SRV _mongodb._tcp.<cluster>.mongodb.net	DNS Test	Checks MongoDB SRV record.

Problems Faced During Setup

1. querySrv ECONNREFUSED

Meaning: Node.js could not resolve the MongoDB SRV record.

Root Cause: DNS resolution issue.

How we debugged: nslookup worked but Node's dns.resolveSrv() failed.

Fix: Forced DNS (Google/Cloudflare) during testing and updated Node.

2. Authentication Failed

Meaning: Atlas accepted the network connection but rejected login.

Root Cause: Username/password mismatch.

Fix: Verified Database User and updated password in .env.

3. Duplicate dotenv Declaration

Reason: server.js content was pasted twice.

Error: Identifier 'dotenv' has already been declared.

Fix: Removed duplicate code.

4. ENOENT package.json

Reason: npm command executed from project root instead of backend folder.

Fix: cd backend then npm run dev.

Important Concepts

Why use .env?

- Protect secrets.
- Different values for development and production.

Why use Mongoose?

- Simplifies CRUD operations.
- Supports schemas and validation.
- Makes MongoDB easier to use from Node.js.

Why process.exit(1)?

- Stops the server if the database is unavailable.

Difference:

Local MongoDB -> mongodb://127.0.0.1:27017

Atlas MongoDB -> mongodb+srv://...

Final Verification

Terminal Output:

MongoDB connected

API running on port 5000

Browser Test:

http://localhost:5000/api/health

Expected Response:

```
{
  "status": "ok",
  "message": "Bitto catalog API is running"
```

}

Interview Questions

What is MongoDB Atlas?

Managed cloud-hosted MongoDB service.

Why use Mongoose instead of MongoDB driver directly?

Provides ODM, schemas, validation and simpler API.

Why use .env?

Keeps secrets outside source code.

What is mongodb+srv?

Uses DNS SRV records to discover MongoDB cluster hosts automatically.

Why whitelist IP?

Atlas blocks unknown IPs for security.